

Exercise_Subquery_MaxAvgDiscount

Tryout > 1: Question 1

Problem Statement

Display the highest average discount (percent) offered on any product as "MAX_DISCOUNT" (column alias).

NOTE: In case you are using subquery, use the same alias names as given in the expected or sample output for inner and outer query.
For the given sample data, the following is the expected output.

Expected Output:

MAX_DISCOUNT
15.20000

Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

Column Name	Data Type and Size	Constraint	Description
VendorId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for each vendor, should start with 'V'
VendorName	VARCHAR(30)	NOT NULL	name of the vendor
VendorContactNo	NUMERIC(10)	UNIQUE	contact number of the vendor

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Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

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VendorId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for each vendor, should start with 'V'
VendorName	VARCHAR(30)	NOT NULL	name of the vendor
VendorContactNo	NUMERIC(10)	UNIQUE	contact number of the vendor
Rating	INTEGER	CHECK	rating of vendor, must be in the range of 0 to 5 (both inclusive)

Table Structure: Product (This table provides the details of all the products)

Column Name	Data Type and Size	Constraint	Description
ProductId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for every product, should start with 'P'
PName	VARCHAR(30)	NOT NULL	name of the product
Brand	VARCHAR(20)	NOT NULL	brand of the product

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Rating	INTEGER	CHECK	rating of vendor, must be in the range of 0 to 5 (both inclusive)
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Table Structure: Product (This table provides the details of all the products)

Column Name	Data Type and Size	Constraint	Description
ProductId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id for every product, should start with 'P'
PName	VARCHAR(30)	NOT NULL	name of the product
Brand	VARCHAR(20)	NOT NULL	brand of the product
PCategory	VARCHAR(20)		category of the product
Color	VARCHAR(10)		color of the product
Review	VARCHAR(10)	CHECK	review of the product, should be 'Bad' or 'Good' or 'Excellent'

Table Structure: Customer (This table provides the details of all the customers)

Column Name	Data Type and Size	Constraint	Description
CustId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id of the customer, must begin with character 'C'

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Table Structure: Customer (This table provides the details of all the customers)

Column Name	Data Type and Size	Constraint	Description
CustId	VARCHAR(5)	PRIMARY KEY, CHECK	unique Id of the customer, must begin with character 'C'
CustName	VARCHAR(10)	NOT NULL	name of the customer
Age	NUMERIC(2)	CHECK	age of the customer, should be between 0 and 99 (both inclusive)
CustLocation	VARCHAR(20)		location, in which the customer resides
Email	VARCHAR(30)	UNIQUE	email id of the customer
Gender	CHAR(1)	CHECK	gender of the customer, should be 'M' or 'F'

Table Structure: VendorOffering (This table provides the details of all the products offered by different vendors)

Column Name	Data Type and Size	Constraint	Description
		FOREIGN	existing VendorId

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Table Structure: VendorOffering (This table provides the details of all the products offered by different vendors)

Column Name	Data Type and Size	Constraint		Description
VendorId	VARCHAR(5)	FOREIGN KEY	PRIMARY KEY	existing VendorId present in Vendor Table
ProductId	VARCHAR(5)	FOREIGN KEY, NOT NULL		existing ProductId present in Product Table
Price	NUMERIC(7,2)	NOT NULL, CHECK		price offered for the product by the vendor in INR, should be greater than zero

Table Structure: Invoice (This table provides the details of all the products purchased by the customers)

Column Name	Data Type and Size	Constraint	Description
InvoiceId	VARCHAR(5)	PRIMARY KEY, CHECK	unique InvoiceId for every Invoice, should start with 'IN'
		FOREIGN KEY	existing CustomerId present in

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Table Structure: Invoice (This table provides the details of all the products purchased by the customers)

Column Name	Data Type and Size	Constraint	Description
InvoiceId	VARCHAR(5)	PRIMARY KEY, CHECK	unique InvoiceId for every Invoice, should start with 'IN'
CustId	VARCHAR(5)	FOREIGN KEY	existing CustId present in Customer table
VendorId	VARCHAR(5)	FOREIGN KEY	existing {VendorId,ProductId} present in VendorOffering table
ProductId	VARCHAR(5)		
QuantityPurchased	NUMERIC(2)	NOT NULL	quantity (number of products) purchased by the customer
Discount	NUMERIC(3,1)	NOT NULL	discount offered on the purchase (in percentage)
PurchaseDate	DATE	NOT NULL	date of purchase

Table Data: Vendor

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Table Data: Vendor

VENDORID	VENDORNAME	VENDORCONTACTNO	RATING
V101	Wendy Creations	8976567890	4
V102	CrazyNuts	NULL	NULL
V103	FlyingHigh Retailers	8126567890	5
V104	Global Traders	8005432556	4
V105	Caretech Enterpises	8116567890	3
V106	Infinity Solutions	8056781234	NULL
V107	BrightWorks Technology	8045678902	3
V108	Walgreens Retailers	8034567891	4

Table Data: Customer

CUSTID	CUSTNAME	AGE	CUSTLOCATION	EMAIL	GENDER
C101	Nancy	23	Plattsburg	nancy@gmail.com	F
C102	Bob	22	Ithaca	bob@gmail.com	M

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V108	Walgreens Retailers	8034567891	4
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Table Data: Customer

CUSTID	CUSTNAME	AGE	CUSTLOCATION	EMAIL	GENDER
C101	Nancy	23	Plattsburg	nancy@gmail.com	F
C102	Bob	33	Ithaca	bob@gmail.com	M
C103	Clara	NULL	Yonkers	clara@gmail.com	F
C104	Vincent	34	Ithaca	vincent@gmail.com	M
C105	Alex	27	Cortland	NULL	M
C106	Peter	77	Yonkers	peter@gmail.com	M
C107	Rose	54	Ithaca	rose@gmail.com	F
C108	Elina	18	Auburn	elina@gmail.com	F

Table Data: Product

PRODUCTID	PNAME	BRAND	PCATEGORY	COLOR	REVIEW
P101	High Boots	Diesel	Shoes	Blue	Good

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Table Data: Product

PRODUCTID	PNAME	BRAND	PCATEGORY	COLOR	REVIEW
P101	High Boots	Diesel	Shoes	Blue	Good
P102	Mens Formal Shirt	Terrain	Clothing	Turquoise	Bad
P103	Pearl Earring	Essence	Accessories	Yellow	Excellent
P104	Kids Crocks	Sparx	NULL	Pink	Excellent
P105	Kids Shirt	Cosmos	Clothing	White	Bad
P106	Sporty Sneakers	Reebok	Shoes	White	Good
P107	Womens Jacket	NorthFace	Clothing	NULL	Good
P108	Leather Wallet	Luxurio	Accessories	Brown	Excellent

Table Data: VendorOffering

VENDORID	PRODUCTID	PRICE
V101	P101	3000.40

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Table Data: VendorOffering

VENDORID	PRODUCTID	PRICE
V101	P101	3000.40
V102	P102	1200.30
V102	P104	1000.80
V103	P101	2800.10
V103	P103	400.20
V104	P104	900.10
V104	P105	500.80
V105	P105	445.00
V107	P107	90.50
V108	P106	699.99

Table Data: Invoice

INVOICEID	CUSTID	VENDORID	PRODUCTID	QUANTITYPURCHASED	DISCOUNT	PU
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V105	P105	445.00
V107	P107	90.50
V108	P106	699.99

Table Data: Invoice

INVOICEID	CUSTID	VENDORID	PRODUCTID	QUANTITYPURCHASED	DISCOUNT	PURCHASEDATE
IN101	C101	V101	P101	2	12.5	2023-08-11
IN102	C102	V102	P102	3	13.0	2023-09-23
IN103	C103	V103	P103	1	15.2	2023-06-10
IN104	C104	V104	P104	1	10.0	2023-09-01
IN105	C103	V105	P105	2	5.0	2023-09-08
IN106	C106	V102	P104	7	10.8	2023-10-02
IN107	C108	V103	P101	4	2.0	2023-10-12
IN108	C107	V104	P105	2	3.3	2023-06-12
IN109	C108	V108	P106	8	7.0	2023-07-18
IN110	C101	V103	P101	10	20.4	2023-08-18

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Problem Statement

Display the maximum number of purchases done by a single customer as "MAX_PURCHASES" (column alias).

NOTE: In case you are using subquery, use the same alias names as given in the expected or sample output for inner and outer query.

For the given sample data, the following is the expected output.

Expected Output:

MAX_PURCHASES
2

Database Structure and Data:

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